

GENERAL PROPERTIES OF MANY PARTICLE SYSTEMS AT LOW TEMPERATURES

NISHTHA TIKALAL

CONTENTS

1 The Dilute Bose Gas

1

1. THE DILUTE BOSE GAS

A simple example of a quantum liquid is a “weakly non-ideal gas.”

Definition 1.1. Weakly non-ideal gas: a gas in which the role of interactions between particles is relatively small.

This regime requires that the scattering amplitude is small compared to the average wavelength $\lambda = \frac{1}{p}$. To ensure that the scattering amplitude is positive, we assume that particle interactions are repulsive. For the Bose gas, this assumption is connected to the fact that even for infinitesimal attractive forces, a Bose gas cannot stay dilute at low temperatures.

Calculating the ground state energy and the energy spectrum for a dilute Bose gas (@ $T = 0$). Assume the particles have spin 0. The interaction energy is of the form:

$$H_{\text{int}} = \frac{U}{2V} \sum_{p_1+p_2=p_3+p_4} a_{p_3}^\dagger a_{p_4}^\dagger a_{p_2} a_{p_1} \quad (1)$$

This should be a familiar equation, as it is the second quantised interaction Hamiltonian. To note, U is the interaction strength, and V is the volume. U is related to the scattering amplitude by:

$$U = \frac{4\pi a}{m} \quad (2)$$

By definition, the s-scattering amplitude is connected with the effective cross-section for the scattering of two identical particles by the formula:

$$d\sigma = (2a)^2 d\Omega,$$

where $d\sigma$ is determined by the Hamiltonian in (1).

In the centre-of-mass frame, standard QM scattering theory dictates that the differential cross-section is given by:

$$d\sigma = \left(\frac{\mu}{2\pi}\right)^2 |M|^2 d\Omega,$$

where μ is the reduced mass.

$$\mu = \frac{m_1 m_2}{m_1 + m_2}.$$

Since the particles are identical, $m_1 = m_2 = m$. $\therefore$ the reduced mass becomes:

$$\mu = \frac{m^2}{2m} = \frac{m}{2}.$$

Note: $M = 2U$. By substitution of M and μ into $d\sigma$, we obtain:

$$d\sigma = \left(\frac{m/2}{2\pi}\right)^2 (2U)^2 d\Omega = \left(\frac{m}{4\pi}\right)^2 (2U)^2 d\Omega$$

By definition, low energy s-wave scattering for identical particles is parameterised by the scattering length a , where the cross section evaluates to:

$$d\sigma = (2a)^2 d\Omega.$$

Setting both forms of $d\sigma$ equal to each other,

$$(2a)^2 d\Omega = \left(\frac{m}{4\pi}\right)^2 (2U)^2 d\Omega$$

$$(2a)^2 = \left(\frac{m}{4\pi} \cdot 2U\right)^2$$

$$2a = \frac{m}{4\pi} \cdot 2U$$

$$a = \frac{mU}{4\pi} \rightarrow U = \frac{4\pi a}{m}$$

To provide some context, Bogoliubov showed that it was possible to diagonalise the Hamiltonian and obtain the energy spectrum. His basic idea was the following: In the ground state, the particles of an ideal Bose gas occupy the lowest level. They have 0 energy and are considered to be in the condensate. In a dilute gas (due to weak interactions), the ground state will differ slightly from that of an ideal gas. The number of particles (N_0) will be greater than the number in other levels, so $N - N_0 \ll N_0$. From this idea, we only consider the following kinds of interactions:

- i) particles in condensate with each other
- ii) "excited" particles with particles in the condensate
- iii) NOT "excited" particles with each other.

The Hamiltonian will only consider the following terms that satisfy the condition:

$$p_1 + p_2 = p_3 + p_4.$$

If $p_1 = p_2 = p_3 = p_4 = 0$,

$$0 + 0 = 0 + 0$$

$$0 = 0//$$

This gives us the term: $a_0^\dagger a_0^\dagger a_0 a_0$. Now, if we let only three of the $p_i = 0$,

$$0 + 0 = 0 + p_4$$

$$0 = p_4$$

$$0 \neq p_4.$$

The initial conservation is not satisfied; any combination of terms associated with this condition are omitted. Moving down the chain, letting two of $p_i = 0$,

$$p_1 + 0 = p_3 + 0$$

$$p_1 = p_3$$

$$p = p \cdot //$$

This tells us that we can have a term with two condensate operators and two excited operators. This gives us the term: $2a_p^\dagger a_0^\dagger a_p a_0$. The 2 accounts for the different permutations that carry the momentum p .

The other permutations are letting:

$$\text{i) } p_1 = p_3 = 0$$

$$\text{ii) } p_1 = p_2 = 0 \rightarrow a_p^\dagger a_{-p}^\dagger a_0 a_0$$

$$\text{iii) } p_3 = p_4 = 0 \rightarrow a_0^\dagger a_0^\dagger a_p a_{-p}$$

Substitution into the Hamiltonian given by (1) yields the result:

$$H_{\text{int}} = \frac{U}{2V} \left[a_0^\dagger a_0^\dagger a_0 a_0 + \sum_{p \neq 0} \left(2a_p^\dagger a_0^\dagger a_p a_0 + 2a_{-p}^\dagger a_0^\dagger a_{-p} a_0 + a_p^\dagger a_{-p}^\dagger a_0 a_0 + a_0^\dagger a_0^\dagger a_p a_{-p} \right) \right] \quad (3)$$

Recall that N_0 is a very large number. We justify regarding the operators $a_0, a_0^\dagger$ as c-numbers, and we replace them by $\sqrt{N_0}$. Updating the Hamiltonian from (3) yields:

$$\begin{aligned} H_{\text{int}} &= \frac{U}{2V} \left[(\sqrt{N_0})^4 + \sum_{p \neq 0} \left(2a_p^\dagger \sqrt{N_0} a_p \sqrt{N_0} + 2a_{-p}^\dagger \sqrt{N_0} a_{-p} \sqrt{N_0} + a_p^\dagger a_{-p}^\dagger (\sqrt{N_0})^2 + (\sqrt{N_0})^2 a_p a_{-p} \right) \right] \\ &= \frac{U}{2V} \left[N_0^2 + 2N_0 \sum_{p \neq 0} (a_{-p}^\dagger a_{-p} + a_p^\dagger a_p) + N_0 \sum_{p \neq 0} (a_p^\dagger a_{-p}^\dagger + a_p a_{-p}) \right] \end{aligned} \quad (4)$$

The total number of particles in the system is written as:

$$N = N_0 + \frac{1}{2} \sum_{p \neq 0} (a_p^\dagger a_p + a_{-p}^\dagger a_{-p}) \quad (5)$$

From (5) we are now able to express N_0 in terms of N , which appears in (4). Re-arranging (5) for N_0 ,

$$N_0 = N - \frac{1}{2} \sum_{p \neq 0} (a_p^\dagger a_p + a_{-p}^\dagger a_{-p})$$

Substitution of this result into (4) yields:

$$H_{\text{int}} = \frac{U}{2V} \left[\left(N - \frac{1}{2} \sum_{p \neq 0} (a_p^\dagger a_p + a_{-p}^\dagger a_{-p}) \right)^2 + 2 \left(N - \frac{1}{2} \sum_{p \neq 0} (a_p^\dagger a_p + a_{-p}^\dagger a_{-p}) \right) \sum_{p \neq 0} (a_{-p}^\dagger a_{-p} + a_p^\dagger a_p) \right]$$

$$+ \left(N - \frac{1}{2} \sum_{p \neq 0} (a_p^\dagger a_p + a_{-p}^\dagger a_{-p}) \right) \sum_{p \neq 0} (a_p^\dagger a_{-p}^\dagger + a_p a_{-p}) \Big]$$

Looking at the first term, it is a difference of squares. Recall the form:

$$(a - b)^2 = a^2 - 2ab + b^2 //$$

Using this result, the first term is expanded, yielding:

$$N^2 - 2N \left(\frac{1}{2} \sum_{p \neq 0} (a_p^\dagger a_p + a_{-p}^\dagger a_{-p}) \right) + b^2.$$

Note that b^2 would contain 4 excited operators, and therefore we can discard it. This leaves us with:

$$N_0^2 \approx N^2 - N \sum_{p \neq 0} (a_p^\dagger a_p + a_{-p}^\dagger a_{-p})$$

Substituting our expression for N_0 into the second term, we obtain:

$$\begin{aligned} & 2 \left(N - \frac{1}{2} \sum_{p \neq 0} (a_p^\dagger a_p + a_{-p}^\dagger a_{-p}) \right) \sum_{p \neq 0} (a_{-p}^\dagger a_{-p} + a_p^\dagger a_p) \\ &= \left(2N - \sum_{p \neq 0} (a_p^\dagger a_p + a_{-p}^\dagger a_{-p}) \right) \sum_{p \neq 0} (a_{-p}^\dagger a_{-p} + a_p^\dagger a_p) \\ &= 2N \sum_{p \neq 0} (a_{-p}^\dagger a_{-p} + a_p^\dagger a_p) \end{aligned}$$

Repeating the process, the third term looks fairly simple:

$$\begin{aligned} & \left(N - \frac{1}{2} \sum_{p \neq 0} (a_p^\dagger a_p + a_{-p}^\dagger a_{-p}) \right) \sum_{p \neq 0} (a_p^\dagger a_{-p}^\dagger + a_p a_{-p}) \\ &= N \sum_{p \neq 0} (a_p^\dagger a_{-p}^\dagger + a_p a_{-p}) \end{aligned}$$

Putting all three expanded terms together, we get:

$$\begin{aligned} H_{\text{int}} = \frac{U}{2V} & \left[N^2 - N \sum_{p \neq 0} (a_p^\dagger a_p + a_{-p}^\dagger a_{-p}) + 2N \sum_{p \neq 0} (a_{-p}^\dagger a_{-p} + a_p^\dagger a_p) \right. \\ & \left. + N \sum_{p \neq 0} (a_p^\dagger a_{-p}^\dagger + a_p a_{-p}) \right] \end{aligned}$$

By distribution and adding the kinetic-energy operator, we obtain:

$$H = \frac{UN^2}{2V} + \left[\frac{1}{2} \sum_{p \neq 0} \left(\frac{p^2}{2m} + \frac{UN}{V} \right) (a_p^\dagger a_p + a_{-p}^\dagger a_{-p}) + \frac{UN}{V} \sum_{p \neq 0} (a_p^\dagger a_{-p}^\dagger + a_p a_{-p}) \right] \quad (6)$$

From (6), the last term is not diagonal. This means that the system can spontaneously scatter, transition, or create/annihilate particles across different states. In order to diagonalise it, we apply a linear transformation of the operators a_p and $a_p^\dagger$:

$$\begin{cases} a_p = \frac{1}{\sqrt{1-A_p^2}}(\alpha_p + A_p\alpha_{-p}^\dagger) \\ a_p^\dagger = \frac{1}{\sqrt{1-A_p^2}}(\alpha_p^\dagger + A_p\alpha_{-p}) \end{cases} \quad (7)$$

From (7), we can note the following:

$$\begin{aligned} a_{-p} &= \frac{1}{\sqrt{1-A_p^2}}(\alpha_{-p} + A_p\alpha_p^\dagger) \\ a_{-p}^\dagger &= \frac{1}{\sqrt{1-A_p^2}}(\alpha_{-p}^\dagger + A_p\alpha_p) \end{aligned}$$

From (7), substitution of these linear transformations into (6) yields:

$$\begin{aligned} H &= \frac{UN^2}{2V} + \left[\frac{1}{2} \sum_{p \neq 0} \left(\frac{p^2}{2m} + \frac{UN}{V} \right) (a_p^\dagger a_p + a_{-p}^\dagger a_{-p}) + \frac{UN}{V} \sum_{p \neq 0} (a_p^\dagger a_{-p}^\dagger + a_p a_{-p}) \right] \\ a_p^\dagger a_p &= \left[\frac{1}{\sqrt{1-A_p^2}}(\alpha_p^\dagger + A_p\alpha_{-p}) \right] \left[\frac{1}{\sqrt{1-A_p^2}}(\alpha_p + A_p\alpha_{-p}^\dagger) \right] \\ &= \frac{\alpha_p^\dagger + A_p\alpha_{-p}}{\sqrt{1-A_p^2}} \cdot \frac{\alpha_p + A_p\alpha_{-p}^\dagger}{\sqrt{1-A_p^2}} \\ &= \frac{(\alpha_p^\dagger + A_p\alpha_{-p})(\alpha_p + A_p\alpha_{-p}^\dagger)}{1-A_p^2} \\ &= \frac{\alpha_p^\dagger\alpha_p + \alpha_p^\dagger A_p\alpha_{-p}^\dagger + A_p\alpha_{-p}\alpha_p + A_p^2\alpha_{-p}\alpha_{-p}^\dagger}{1-A_p^2} \end{aligned}$$

Notice how the last term is in the wrong order. Let $\alpha_{-p}\alpha_{-p}^\dagger = 1 + \alpha_{-p}^\dagger\alpha_{-p}$. Thus,

$$a_p^\dagger a_p = \frac{A_p^2 + \alpha_p^\dagger\alpha_p + A_p^2\alpha_{-p}^\dagger\alpha_{-p} + A_p(\alpha_p^\dagger\alpha_{-p}^\dagger + \alpha_{-p}\alpha_p)}{1-A_p^2}$$

We repeat the product expansion for $a_{-p}^\dagger a_{-p}$, with the exception of a sign change ($p \rightarrow -p$). This yields:

$$a_{-p}^\dagger a_{-p} = \frac{A_p^2 + \alpha_{-p}^\dagger\alpha_{-p} + A_p^2\alpha_p^\dagger\alpha_p + A_p(\alpha_{-p}^\dagger\alpha_p^\dagger + \alpha_p\alpha_{-p})}{1-A_p^2}$$

Notice how the last term is in the wrong order. Let $\alpha_{-p}\alpha_{-p}^\dagger = 1 + \alpha_{-p}^\dagger\alpha_{-p}$. Thus,

$$a_p^\dagger a_p = \frac{A_p^2 + \alpha_p^\dagger\alpha_p + A_p^2\alpha_{-p}^\dagger\alpha_{-p} + A_p(\alpha_p^\dagger\alpha_{-p}^\dagger + \alpha_{-p}\alpha_p)}{1-A_p^2}$$

We repeat the product expansion for $a_{-p}^\dagger a_{-p}$, with the exception of a sign change ($p \rightarrow -p$). This yields:

$$a_{-p}^\dagger a_{-p} = \frac{A_p^2 + \alpha_{-p}^\dagger \alpha_{-p} + A_p^2 \alpha_p^\dagger \alpha_p + A_p(\alpha_{-p}^\dagger \alpha_p^\dagger + \alpha_p \alpha_{-p})}{1 - A_p^2}$$

Combining $a_p^\dagger a_p$ and $a_{-p}^\dagger a_{-p}$ yields:

$$\frac{1}{1 - A_p^2} \left[2A_p^2 + (1 + A_p^2)(\alpha_p^\dagger \alpha_p + \alpha_{-p}^\dagger \alpha_{-p}) + 2A_p(\alpha_p^\dagger \alpha_{-p}^\dagger + \alpha_{-p} \alpha_p) \right]$$

Substitution of this final result into (6) yields:

$$\begin{aligned} H = & \frac{UN^2}{2V} + \sum_{p \neq 0} \frac{1}{1 - A_p^2} \left[\left(\frac{p^2}{2m} + \frac{UN}{V} \right) A_p^2 + \frac{UN}{V} A_p \right] \\ & + \frac{1}{2} \sum_{p \neq 0} \frac{1}{1 - A_p^2} \left[\left(\frac{p^2}{2m} + \frac{UN}{V} \right) (1 + A_p^2) + \frac{2UN}{V} A_p \right] (\alpha_p^\dagger \alpha_p + \alpha_{-p}^\dagger \alpha_{-p}) \\ & + \frac{1}{2} \sum_{p \neq 0} \frac{1}{1 - A_p^2} \left[\left(\frac{p^2}{2m} + \frac{UN}{V} \right) 2A_p + \frac{UN}{V} (1 + A_p^2) \right] (\alpha_p^\dagger \alpha_{-p}^\dagger + \alpha_{-p} \alpha_p) \end{aligned} \quad (8)$$

Since the last term in (8) is the non-diagonal one, for it to vanish, we require that:

$$\left(\frac{p^2}{2m} + \frac{UN}{V} \right) 2A_p + \frac{UN}{V} (1 + A_p^2) = 0$$

Solving for A_p yields:

$$\left(\frac{p^2}{2m} + \frac{UN}{V} \right) 2A_p + \frac{UN}{V} + \frac{UN}{V} A_p^2 = 0$$

$$\frac{UN}{V} A_p^2 + \left(\frac{p^2}{m} + \frac{2UN}{V} \right) A_p + \frac{UN}{V} = 0 ; \text{ which we know as a quadratic equation :}$$

$$a = UN/V$$

$$b = p^2/m + 2UN/V$$

$$c = UN/V$$

$$\begin{aligned} A_p = & \frac{-\left(\frac{p^2}{m} + \frac{2UN}{V}\right) \pm \sqrt{\left(\frac{p^2}{m} + \frac{2UN}{V}\right)^2 - 4\left(\frac{UN}{V}\right)\left(\frac{UN}{V}\right)}}{2\left(\frac{UN}{V}\right)} \\ = & \frac{V}{UN} \left[-\frac{p^2}{2m} - \frac{UN}{V} \pm \sqrt{\left(\frac{p^2}{2m} + \frac{UN}{V}\right)^2 - \left(\frac{UN}{V}\right)^2} \right] \end{aligned} \quad (9)$$

Substituting our value of A_p from (9) into (8) yields:

$$H = \frac{UN^2}{2V} - \frac{1}{2} \sum_{p \neq 0} \left[\left(\frac{p^2}{2m} + \frac{UN}{V} \right) - \sqrt{\left(\frac{p^2}{2m} + \frac{UN}{V} \right)^2 - \left(\frac{UN}{V} \right)^2} \right] + \frac{1}{2} \sum_{p \neq 0} \sqrt{\left(\frac{p^2}{2m} + \frac{UN}{V} \right)^2 - \left(\frac{UN}{V} \right)^2} (\alpha_p^\dagger \alpha_p + \alpha_{-p}^\dagger \alpha_{-p}) \quad (10)$$

We note that the first two terms are a constant, but the third term can be re-expressed as a diagonal operator:

$$\sum_{p \neq 0} n_p \varepsilon(\vec{p}),$$

where n_p are the occupation numbers of the operators α_p . The smallest value is obtained when $n_p = 0$. This, (10) is the excitation energy.

The weakly excited states of a dilute Bose gas can be described with an energy spectrum:

$$\varepsilon(\vec{p}) = \sqrt{\left(\frac{p^2}{2m} + \frac{UN}{V}\right)^2 - \left(\frac{UN}{V}\right)^2} \quad (11)$$

In the limit of small momenta,

$$\varepsilon(\vec{p}) \approx \sqrt{\frac{4\pi a N/V}{m}} p, \quad (12)$$

which corresponds to the phonon part of the spectrum of a Bose liquid. For large momenta,

$$\varepsilon(\vec{p}) \approx \frac{p^2}{2m}, \quad (13)$$

which we know to be the energy of a free particle.

It is essential that the scattering amplitude a should be small and finite, which makes expansion of energy with respect to a possible. The relation between U and the scattering amplitude is not exact, as seen by (2). To correct it, we use second-order perturbation theory to examine the scattering of two particles of a transition to the states $\vec{p}, -\vec{p}$.

Recall by (2) $U = \frac{4\pi a}{m}$.

$$U - \frac{U^2}{V} \sum_{p \neq 0} \frac{1}{p^2/m} = \frac{4\pi a}{m} \quad (14)$$

Expressing U in terms of a ,

$$\begin{aligned} U &= \frac{4\pi a}{m} + \frac{U^2}{V} \sum_{p \neq 0} \frac{1}{p^2/m} \\ &= \frac{4\pi a}{m} + \frac{\left(\frac{4\pi a}{m}\right)^2}{V} \sum_{p \neq 0} \frac{1}{p^2/m} \\ &= \frac{4\pi a}{m} + \frac{1}{V} \left(\frac{4\pi a}{m}\right)^2 \sum_{p \neq 0} \frac{1}{p^2/m} \end{aligned}$$

Substitution of this result into (10) yields the energy at the ground state:

$$\begin{aligned} E &= \frac{2\pi a}{m} \frac{N^2}{V} + \frac{8\pi^2 a^2}{m^2} \left(\frac{N}{V}\right)^2 \sum_{p \neq 0} \frac{1}{p^2/m} - \frac{1}{2} \sum_{p \neq 0} \left(\frac{p^2}{2m} + \frac{4\pi a N}{mV}\right) \\ &\quad \cdot \left[1 - \sqrt{1 - \left(\frac{4\pi a N/mV}{\frac{p^2}{2m} + \frac{4\pi a N}{mV}}\right)^2} \right] \end{aligned} \quad (15)$$

Integrating over the momenta, let:

$$\begin{cases} \epsilon_p = \frac{p^2}{2m} \\ \Delta = 4\pi a N/mV \end{cases}$$

With this, (15) becomes:

$$\frac{\Delta^2}{4\epsilon_p} - \frac{1}{2}(\epsilon_p + \Delta) \left[1 - \sqrt{1 - \left(\frac{\Delta}{\epsilon_p + \Delta} \right)^2} \right]$$

Simplifying the inside of the radical:

$$1 - \left(\frac{\Delta}{\epsilon_p + \Delta} \right)^2 = \frac{(\epsilon_p + \Delta)^2 - \Delta^2}{(\epsilon_p + \Delta)^2} = \frac{\epsilon_p^2 + 2\epsilon_p\Delta}{(\epsilon_p + \Delta)^2}$$

Again, (15) becomes:

$$\frac{\Delta^2}{4\epsilon_p} - \frac{1}{2}(\epsilon_p + \Delta) + \frac{1}{2}\sqrt{\epsilon_p^2 + 2\epsilon_p\Delta}$$

In order to solve this, we define the following:

$$\begin{aligned} \epsilon_p &= \Delta x^2 \rightarrow \frac{p^2}{2m} = \Delta x^2 \\ p &= \sqrt{2m\Delta} x \end{aligned}$$

Substitution into (15) yields:

$$\begin{aligned} \Delta \left[\frac{1}{4x^2} - \frac{1}{2}(x^2 + 1) + \frac{1}{2}\sqrt{x^4 + 2x^2} \right] \\ = \frac{\Delta}{2} \left[\frac{1}{2x^2} - (x^2 + 1) + \sqrt{x^4 + 2x^2} \right] \end{aligned}$$

This will be the function we integrate. Using spherical coordinates,

$$\frac{V}{(2\pi)^3} \int_0^\infty 4\pi p^2 dp = \frac{V}{2\pi^2} \int_0^\infty p^2 dp$$

from $p = \sqrt{2m\Delta} x$, $dp = \sqrt{2m\Delta} dx$. Thus, the integral becomes:

$$\begin{aligned} &= \frac{V}{2\pi^2} \int_0^\infty (\sqrt{2m\Delta} x)^2 \sqrt{2m\Delta} dx \\ &= \frac{V}{2\pi^2} \int_0^\infty (2m\Delta)^{3/2} x^2 dx \\ &= \frac{V}{2\pi^2} (2m\Delta)^{3/2} \frac{\Delta}{2} \int_0^\infty x^2 \left[\frac{1}{2x^2} - (x^2 + 1) + \sqrt{x^4 + 2x^2} \right] dx \end{aligned}$$

This models relations from hyperbolic trigonometry.

$$= \frac{V}{2\pi^2} (2m\Delta)^{3/2} \frac{\Delta}{2} \left(\frac{\sqrt{2}}{15} \right)$$

$$\therefore \frac{E}{V} = \frac{2\pi a}{m} \left(\frac{N}{V} \right)^2 \left[1 + \frac{128}{15\sqrt{\pi}} a^{3/2} \left(\frac{N}{V} \right)^{1/2} \right] \quad (16)$$

From (16), we note that the velocity of sound is:

$$u = \sqrt{\frac{V^2}{mN} \frac{\partial^2 E}{\partial V^2}} = \sqrt{\frac{4\pi a N / m V}{m}} \quad (17)$$

The momentum distribution of the excitations is given by the Bose formula:

$$\bar{n}_p = \alpha_p^\dagger \alpha_p = \frac{1}{e^{\varepsilon(p)/T} - 1} \quad (18)$$

The momentum distribution of the particles themselves in a Bose liquid can be found by:

$$N_p = a_p^\dagger a_p$$

By (7), recall that:

$$\begin{cases} a_p = \frac{1}{\sqrt{1-A_p^2}} (\alpha_p + A_p \alpha_{-p}^\dagger) \\ a_p^\dagger = \frac{1}{\sqrt{1-A_p^2}} (\alpha_p^\dagger + A_p \alpha_{-p}) \end{cases}$$

Substitution into the expression above yields:

$$\begin{aligned} \bar{N}_p &= \frac{1}{\sqrt{1-A_p^2}} (\alpha_p + A_p \alpha_{-p}^\dagger) \frac{1}{\sqrt{1-A_p^2}} (\alpha_p^\dagger + A_p \alpha_{-p}) \\ &= \frac{1}{1-A_p^2} (\alpha_p + A_p \alpha_{-p}^\dagger) (\alpha_p^\dagger + A_p \alpha_{-p}) \\ &= \frac{1}{1-A_p^2} (\alpha_p \alpha_p^\dagger + A_p \alpha_p \alpha_{-p} + A_p \alpha_{-p}^\dagger \alpha_p^\dagger + A_p^2 \alpha_{-p}^\dagger \alpha_{-p}) \\ &= \frac{1}{1-A_p^2} (\alpha_p^\dagger \alpha_p + A_p^2 \alpha_{-p} \alpha_{-p}^\dagger) \\ &= \frac{\bar{n}_p + A_p^2 (\bar{n}_p + 1)}{1-A_p^2} \end{aligned} \quad (19)$$

The number of particles of zero energy is obtained from the formula:

$$N_0 = N - \sum_{p \neq 0} \bar{N}_p$$

At absolute zero, $\bar{n}_p = 0$, so that by (19):

$$\bar{N}_p = \frac{\frac{8\pi^2 a^2}{m^2} \left(\frac{N}{V}\right)^2}{\varepsilon(p) \left[\varepsilon(p) + \frac{p^2}{2m} + \frac{4\pi a N}{mV} \right]} \quad (20)$$

$$\frac{N_0}{N} = 1 - \frac{8}{3\sqrt{\pi}} a^{3/2} \left(\frac{N}{V}\right)^{1/2} \quad (21)$$

Notably, even in the ground state, in a non-ideal Bose gas, not all particles have zero momentum.

REFERENCES

- [1] D. J. Griffiths and D. F. Schroeter, *Introduction to Quantum Mechanics*, 3rd ed. Cambridge, UK: Cambridge University Press, 2018.
- [2] A. A. Abrikosov, L. P. Gorkov, and I. E. Dzyaloshinski, *Methods of Quantum Field Theory in Statistical Physics*. New York, NY: Dover Publications, 1975.
- [3] S. J. Blundell and K. M. Blundell, *Concepts in Thermal Physics*, 2nd ed. Oxford, UK: Oxford University Press, 2009.